Report R22 — what the transition graph means

Weak components, terminal components and Hilbert-space fragmentation in open quantum systems

Tier 1 analysis

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Abstract

Every sector count this project reports is a count of components of one directed graph: the support graph of the channel in the computational basis. For unitary rules the choice of component notion is immaterial, and the project has been able to say "Krylov sector" without qualification. For dissipative rules it is not immaterial, and the two natural counts can disagree *exponentially*: we have rules with a single weakly connected component and ψ^N terminal components, $\psi = 1.4656$. This report says what the two counts are in the language of open quantum systems, and answers the question that provoked it — if the recurrent classes proliferate but the weak components do not, is that Hilbert-space fragmentation? The answer is **no**, and the reason is sharp rather than terminological. Weak components are exactly the blocks of a *strong symmetry* in the sense of Buča and Prosen: the finest decomposition of the basis that makes every Kraus operator block-diagonal, hence a genuine superselection rule with a charge measurable at $t = 0$. Terminal components are the *enclosures* of the Baumgartner–Narnhofer / Albert–Jiang asymptotic decomposition: they count steady states, not sectors. Made dimensional, the two counts are $\dim(\mathcal{C} \cap \mathcal{D}) = n_{\mathrm{wcc}}$ and $\dim(\mathcal{F} \cap \mathcal{D}) = n_{\mathrm{rec}}$ for the strong-symmetry commutant $\mathcal{C}$ and the conserved-quantity space $\mathcal{F}$, so the graph inequality $n_{\mathrm{rec}} \geq n_{\mathrm{wcc}}$ is just the inclusion $\mathcal{C} \subseteq \mathcal{F}$, and Moudgalya–Motrunich's commutant criterion applies verbatim to the first. The central result is that the gap between them is invisible to the *idempotents*: the projections in $\mathcal{F}$ are exactly the indicators of unions of weak components, so a channel may acquire ψ^N steady states without the Boolean algebra of always-answerable questions gaining a single atom. Fragmentation is exponential growth of that Boolean algebra; exponentially many attractors inside one block is multistability. Two hypotheses bound when the gap can occur at all: unital channels have no transient part (so all three component notions coincide), and functional support graphs have one attractor per component — **the gap requires the channel to be simultaneously non-unital and branching**, which 891 measured points across the 81 deterministic rules confirm without exception. Operationally, a strong symmetry says the attractor is a deterministic function of the initial state: measured, 100% of basis states have a certain fate for rules whose sectors proliferate, against 46% at $N = 10$ and 39% at $N = 12$ for a rule with one weak component. Yet the retained information $I(X;T)$ is extensive in *both* regimes, at comparable rates — so the difference is not how much is remembered but with what certainty: fragmentation is a noiseless code, multistability a noisy channel of similar capacity from which nothing at all can be recovered with probability one. We map the four growth regimes onto the literature, flag a terminology collision between two independent "strong/weak" dichotomies that this project uses both of, and state precisely what a population-level graph can and cannot certify. Finally we lift the main such limitation: the asymptotic manifold $\bigoplus_k \rho_k \otimes \sigma_k$ is computed exactly for all 256 rules at $N = 4$ and 5, splitting each enclosure into the decoherence-free factor n_k that holds qubits forever and the mixing factor m_k that is erased. No block anywhere has both factors non-trivial, and two rules this report had filed as multistable turn out to have a *single* coherent block — for W_{203} the multistability belongs to the monitored chain rather than to the channel, while W_{36} keeps a genuine coherent gap. Read as codes, the three counts separate cleanly: n_{wcc} is zero-error classical capacity, n_k is protected qubits, and $n_{\mathrm{rec}} - n_{\mathrm{wcc}}$ is nothing. That reading has a sharp consequence for autonomous error correction, which we test on the rule Guedes,

Winter and Müller propose for it: W_{232} is majority voting, its asymptotic algebra is the full $M_{n_{\mathrm{rec}}}$, and exactly because everything is protected it corrects one error at every system size while its codeword basin decays like 0.777^N. Fragmentation makes an excellent passive memory and a bad decoder.

Contents

1 Why the question does not arise for closed systems

The object under study is the support graph G of one full brick-wall cycle Φ in the computational basis: vertices are the 2^N bit strings, and $x \to y$ when the channel can take $|x\rangle\langle x|$ to something with weight on $|y\rangle\langle y|$. Writing $\Phi(\rho) = \sum_j K_j \rho K_j^\dagger$, this is

$$x \to y \quad \Longleftrightarrow \quad \exists j : \langle y|K_j|x\rangle \neq 0. \tag{1}$$

Three notions of "component" live on a directed graph: weakly connected components (WCCs, obtained by forgetting edge directions), strongly connected components (SCCs, mutual reachability), and terminal SCCs — those with no outgoing edge, which we also call recurrent classes or attractors. In general these are different. The following says why the distinction has never had to be made in the closed-system fragmentation literature.

Throughout, P denotes the induced population dynamics,

$$P_{yx} \;=\; \sum_j \big|\langle y|K_j|x\rangle\big|^2, \tag{2}$$

a column-stochastic matrix whose support is exactly G. Two structural hypotheses turn out to control everything, and neither is the one we first reached for.

Proposition 1 (Unitality kills the transient part)**.** *If Φ is* unital*, $\Phi(\mathbb{1}) = \mathbb{1}$, then P is doubly stochastic, no state is transient, and*

$$\textit{WCC} \;=\; \textit{SCC} \;=\; \textit{terminal SCC}.$$

Unitary evolution is the special case $\Phi(\rho) = U\rho U^\dagger$.

Proof. Columns of P sum to one by trace preservation, $\sum_j K_j^\dagger K_j = \mathbb{1}$. For the rows, $\sum_x P_{yx} = \sum_j \langle y|K_j K_j^\dagger|y\rangle = \langle y|\Phi(\mathbb{1})|y\rangle$, which is 1 exactly when Φ is unital on the diagonal. So P is doubly stochastic, the uniform distribution is stationary with full support, and since a transient state of a finite chain carries zero stationary mass, there are none; every SCC is therefore terminal. Finally, distinct terminal SCCs admit no edge in either direction — an edge leaving a terminal SCC contradicts terminality — so each is its own WCC. □

Stating it for unital rather than unitary channels is not generality for its own sake: it identifies the physical ingredient. Unitality says the maximally mixed state is stationary, i.e. the channel has no contracting direction in state space. A Hadamard branches but does not contract; a reset contracts. That is why the phenomenon of this report is invisible until a D or an E enters the rule word. Empirically the dichotomy is exact: across all 16 unitary rules and all $N \in [6, 16]$ the measured transient fraction is identically 0 and $n_{\text{rec}} = n_{\text{wcc}}$ at every point, while no dissipative rule has an empty transient part at any N; measured row sums of P deviate from 1 by as much as 15 for W_{36} at $N = 8$.

Contraction alone is not enough, however. The second hypothesis is branching.

Proposition 2 (Determinism also collapses the two counts)**.** *If every vertex of G has out-degree one — the support graph is a* functional *graph — then every WCC contains exactly one terminal SCC, so $n_{\text{rec}} = n_{\text{wcc}}$ regardless of how contracting Φ is.*

Proof. A weakly connected component of a functional graph on n vertices has exactly n edges, hence exactly one cycle; following the unique successor from any vertex must reach it. That cycle is the unique terminal SCC of the component. □

Together the two propositions bound the phenomenon from both sides:

> **A gap between n_{wcc} and n_{rec} requires the channel to be simultaneously non-unital and branching.**

Both halves are visible in our rule family. The 81 rules whose words contain no V have out-degree one, and across all 891 measured (rule, N) points among them $n_{\text{rec}} = n_{\text{wcc}}$ without a single exception. The 16 rules with no D or E are unital, and satisfy the same equality by Proposition 1. Every rule for which the two counts actually separate therefore carries at least one V *and* at least one reset — and, as it happens, all eight carry exactly one V, the minimum amount of branching that permits the gap at all. **The subject of this report is what happens when branching and contraction act together, and it has no closed-system shadow.**

2 The dictionary

2.1 Weak components are a strong symmetry

Proposition 3 (WCCs block-diagonalise every Kraus operator)**.** *Let $\{B_\alpha\}$ be the WCCs of G and P_α the orthogonal projector onto* $\mathrm{span}\{|x\rangle : x \in B_\alpha\}$. *Then $[K_j, P_\alpha] = 0$ for every Kraus operator K_j and every α, and consequently $\mathrm{Tr}(P_\alpha \rho)$ is conserved for all time. Conversely, the $\{B_\alpha\}$ are the* finest *partition of the computational basis with this property.*

Proof. If $P_\alpha K_j P_\beta \neq 0$ for $\alpha \neq \beta$ there are $x \in B_\beta$, $y \in B_\alpha$ with $\langle y|K_j|x\rangle \neq 0$, i.e. an edge $x \to y$ in (1); then x and y are weakly connected and $\alpha = \beta$, a contradiction. Hence each $K_j = \bigoplus_\alpha P_\alpha K_j P_\alpha$ and $P_\alpha K_j = K_j P_\alpha$. Conservation follows: $\mathrm{Tr}(P_\alpha \Phi(\rho)) = \sum_j \mathrm{Tr}(K_j P_\alpha \rho K_j^\dagger) = \sum_j \mathrm{Tr}(K_j^\dagger K_j P_\alpha \rho) = \mathrm{Tr}(P_\alpha \rho)$, using $\sum_j K_j^\dagger K_j = \mathbb{1}$. Minimality holds because any coarser-than-G partition with block-diagonal Kraus operators has no edges between its parts, so it is a union of WCCs. □

Proposition 3 is the whole content of the identification. In the classification of Buča and Prosen [1], a symmetry of a Lindbladian is *strong* when the Hamiltonian *and every jump operator individually* commute with it, and *weak* when only the generator as a whole does. Setting $U = \sum_\alpha e^{i\theta_\alpha} P_\alpha$ for generic phases, Proposition 3 says exactly that the WCC decomposition furnishes a strong symmetry, and the minimality clause says it is the maximal one available in the computational basis. Buča and Prosen establish the fact that gives this report its answer: *only* a strong symmetry implies non-trivial reductions inside the space of steady states.

The statement can be made dimensional, which is what connects it to the commutant criterion. Write $\mathcal{C} = \{A : [A, K_j] = 0 \ \forall j\}$ for the commutant of the Kraus set — the algebra of strong symmetries — and $\mathcal{D}$ for the algebra of operators diagonal in the computational basis.

Proposition 4 (n_{wcc} *is* a commutant dimension)**.** $\dim(\mathcal{C} \cap \mathcal{D}) = n_{\mathrm{wcc}}$. *Concretely, $\mathcal{C} \cap \mathcal{D}$ consists exactly of the operators $A = \sum_x a(x)|x\rangle\langle x|$ whose symbol a is constant on each weak component.*

Proof. For diagonal A, $[A, K_j] = 0$ reads $(a(y) - a(x))\,\langle y|K_j|x\rangle = 0$ for all x, y, j, i.e. $a(y) = a(x)$ whenever $x \to y$. Since a is a scalar function, the condition is symmetric under reversing the edge, so a is constant on weakly connected components — and any such a manifestly commutes with every K_j. □

This is the precise bridge to Moudgalya and Motrunich [5], whose criterion for fragmentation is exponential growth of the dimension of the commutant algebra, with conventional symmetries such as $U(1)$ or $SU(2)$ giving only polynomial growth. Proposition 4 says that n_{wcc} is not merely *correlated* with that dimension; in the classical (product-basis) setting it *is* that dimension. So:

> n_{wcc} counts superselection sectors, and equals the dimension of the diagonal strong-symmetry commutant. It is the open-system successor of "number of Krylov sectors", and it is the count that fragmentation is about.

2.2 Terminal components are enclosures

The terminal SCCs are a different kind of object entirely. They are the *enclosures* of Baumgartner and Narnhofer [2], whose analysis of GKS–Lindblad generators equips the Hilbert space with a decomposition into mutually orthogonal subspaces and separates the transient part from the part where probability accumulates. In the quantum-channel formulation the same structure is the irreducible decomposition of Carbone and Pautrat [4]: the recurrent subspace, which supports the invariant states, decomposes into the supports of the extremal invariant states, with the orthogonal complement being transient. Albert and Jiang [3] give the form of the asymptotic manifold,

$$\rho(\infty) = \bigoplus_k (\sigma_k \otimes \rho_k), \qquad \sigma_k \in M_{n_k}, \tag{3}$$

where k runs over the enclosures, ρ_k is a fixed state on the "dissipative" factor of enclosure k, and M_{n_k} is a noiseless subsystem carrying surviving coherence. Their central point — and the one we lean on in §3 — is that the infinite-time state is fixed by the initial state together with the full list of conserved quantities.

So:

> n_{rec} counts steady states. It is a statement about the $t \to \infty$ limit, not about a decomposition of the Hilbert space.

2.3 The table

graph object	algebraic object	physical reading
WCC	block of a strong symmetry; $[K_j, P_\alpha] = 0$	superselection sector; Krylov sector of the dephased model
n_{wcc}	# blocks	# conserved charges valid at *all* times
$d^{\text{wcc}}_{\max}$	largest block dimension	the R21 strong/weak-fragmentation ratio
SCC	communicating class	refines the sector into transient strata
transient SCCs	transient subspace	everything that decays
terminal SCC	enclosure [2]; irreducible component [4]	attractor; support of one extremal steady state
n_{rec}	# enclosures	# extremal steady states
$d^{\text{rec}}_{\max}$	$n_k \cdot m_k$ of the largest enclosure	noiseless-subsystem dimension *times* mixing dimension

Two structural facts follow immediately and are worth stating because they constrain every census. First, $n_{\text{rec}} \geq n_{\text{wcc}}$ always, since every weak component contains at least one terminal SCC; the inequality is saturated exactly when each sector has a unique attractor. Second, the familiar hyperbola $ab \geq 2$ of R9/R16 is a statement about WCCs *only*, and it holds because they tile: $\sum_\alpha \dim B_\alpha = 2^N$. Recurrent classes do not tile — they occupy the recurrent volume $\mu^N \leq 2^N$ — so the correct analogue is $a_{\text{rec}}\, b_{\text{rec}} \geq \mu$, with $\mu = 2$ recovering the closed-system bound. For the rules discussed below $\mu = \psi = 1.4656$ and the bound is saturated exactly.

3 Is exponential n_{rec} without exponential n_{wcc} fragmentation?

3.1 The criterion

The sharpest available definition of fragmentation is the commutant-algebra one of Moudgalya and Motrunich [5]: given the set of local terms (Hamiltonian terms, or the local gates of a circuit), form the algebra of all operators commuting with each of them; the system is fragmented when the dimension of this *commutant* grows exponentially in system size, as against the polynomial growth produced by conventional symmetries such as $U(1)$ or $SU(2)$. Their framework also supplies the classical/quantum distinction — "classical" fragmentation being fragmentation in a product basis, which is the case here.

The open-system version replaces "commutes with each local term" by "commutes with the Hamiltonian and each jump operator", which is Buča–Prosen strong symmetry, and the superoperator formulation is developed in [6], where symmetry algebras appear as frustration-free ground states of a local super-Hamiltonian and the strong/weak distinction is made explicit. Under that criterion the answer is immediate: Proposition 3 identifies the strong-symmetry commutant in the computational basis with the WCC decomposition, so **exponentially many weak components is fragmentation and exponentially many terminal components is not**. A rule with one weak component has a trivial strong-symmetry commutant: its Hilbert space is a single Krylov sector.

3.2 What is exponentially large instead

This does not make the phenomenon empty — something is exponentially large, and it is worth naming precisely. It is the space of conserved quantities in the Heisenberg picture. For each terminal component T_i define the *absorption probability*

$$\varphi_i(x) \;=\; \Pr[\,\text{absorbed in } T_i \mid \text{start in } x\,], \qquad J_i \;=\; \sum_x \varphi_i(x)\,|x\rangle\langle x|. \tag{4}$$

Each J_i satisfies $\Phi^\dagger(J_i) = J_i$, so $\mathrm{Tr}(J_i\rho(t))$ is constant: these are conserved quantities in exactly Albert and Jiang's sense. Writing $\mathcal{F} = \{A : \Phi^\dagger(A) = A\}$ for the space of them, the counterpart of Proposition 4 holds.

Proposition 5 (n_{rec} is also a dimension, of a larger space)**.** *For the* monitored *channel — the one whose Heisenberg dual is the population chain —* $\dim(\mathcal{F}\cap\mathcal{D}) = n_{\mathrm{rec}}$*, with basis* $\{J_i\}$*. Moreover* $\mathcal{C} \subseteq \mathcal{F}$*, so* $n_{\mathrm{wcc}} \le n_{\mathrm{rec}}$ *always.*

Proof. A diagonal A with symbol a is fixed by the population chain's dual iff a is harmonic, $a(x) = \sum_y P_{yx}\, a(y)$. By the maximum principle a is constant on each terminal component, and its values there determine it everywhere by solving the Dirichlet problem on the transient part; the solutions are spanned by the φ_i. The inclusion is immediate: $[A, K_j] = 0$ gives $\Phi^\dagger(A) = \sum_j K_j^\dagger A K_j = A \sum_j K_j^\dagger K_j = A$. □

Remark 1 (The unmonitored channel gives only an inequality)*.* The restriction to the monitored channel in Proposition 5 is not cosmetic. For diagonal A the *diagonal* of $\Phi^\dagger(A)$ is exactly the population chain's answer, $\langle x|\Phi^\dagger(A)|x\rangle = \sum_y P_{yx} a(y)$, so harmonicity is necessary — but $\Phi^\dagger(A)$ need not itself be diagonal, and $A \in \mathcal{F}\cap\mathcal{D}$ requires the off-diagonal part to vanish as well. Hence, for the coherent channel,

$$n_{\mathrm{wcc}} \;=\; \dim(\mathcal{C}\cap\mathcal{D}) \;\le\; \dim(\mathcal{F}\cap\mathcal{D}) \;\le\; n_{\mathrm{rec}},$$

and the right-hand inequality can be *strict*: W_{29} at $N = 4$ has three terminal components and only two conserved diagonal quantities. R24 measures the squeeze across this family and finds that several of the gap rules of §4 owe their gap entirely to the dephasing — while W_{36}, W_{44}, W_{100} keep a genuine coherent gap at $N = 5$. Everything downstream of Proposition 5 in this report is a statement about the monitored channel, which is the object §3.3 and §3.4 measure; Theorem 1 is unaffected, since it concerns projections and $\mathcal{C}$ and $\mathcal{F}$ share those (Prop. 6).

So the graph inequality $n_{\mathrm{rec}} \ge n_{\mathrm{wcc}}$ is nothing but the algebra inclusion $\mathcal{C} \subseteq \mathcal{F}$: *every strong symmetry is a conserved quantity, but not every conserved quantity is a strong symmetry.* Exponentially many attractors therefore do imply exponentially many conserved quantities. The question is what *kind* — and the answer is that the two spaces differ in a way no amount of multistability can repair.

Theorem 1 (The idempotents never proliferate)**.** *The projections lying in* $\mathcal{F}\cap\mathcal{D}$ *are exactly the indicators of unions of weak components. Hence* $\mathcal{F}\cap\mathcal{D}$ *contains precisely* $2^{n_{\mathrm{wcc}}}$ *projections, however large* n_{rec} *may be, and its minimal projections are the* P_α *of Proposition 3. In particular* $J_i^2 = J_i$ *if and only if* T_i *is the unique terminal component of its weak component, and* $J_i^2 = J_i$ *for* every i *if and only if* $n_{\mathrm{rec}} = n_{\mathrm{wcc}}$*.*

Proof. Let $Q = \sum_{x \in S} |x\rangle\langle x|$ be harmonic. For $x \in S$, $1 = \sum_{y \in S} P_{yx}$ forces every successor of x into S; for $x \notin S$, $0 = \sum_{y \in S} P_{yx}$ forbids any successor of x from lying in S. Thus S and its complement are both forward-closed, so no edge joins them in either direction and S is a union of weak components. Conversely every such indicator is harmonic. The count and the statement about minimal projections follow, and the J_i clause is the special case $S = \operatorname{supp} \varphi_i$. □

Theorem 1 is the whole answer, and it is stronger than the observation that non-idempotent conserved quantities exist. *The idempotent part of the conserved-quantity space is blind to the attractors entirely.* A rule may acquire ψ^N steady states, and the Boolean algebra of yes/no questions about the initial state that remain answerable at all times does not grow by a single atom. Fragmentation is exponential growth of that Boolean algebra; multistability is exponential growth of the surrounding linear space and nothing more. Measured directly at $N = 8$: W_{73} has 19 idempotent J_i out of 19, W_{36} has 9 out of 28 — one for each of the nine weak components that happen to hold a single attractor — and W_{203}, whose entire state space is one weak component, has *none*.

3.3 The operational content, measured

Proposition 1 converts the abstract distinction into a number one can compute: what fraction of basis states have their attractor determined *with certainty* by the initial condition? We built the honest population Markov chain of the channel (I keeps, $V = H$ splits $\frac{1}{2}/\frac{1}{2}$, D and E reset) and solved $\varphi = Q\varphi + R$ exactly on the transient block.

rule	regime	n_{wcc}	n_{rec}	certain fate	effective # attractors
W_{73} `VIID`	sectors proliferate	41	41	100.0%	1.00
W_{109} `EIIV`	sectors proliferate	54	54	100.0%	1.00
W_{36} `IDDV`	attractors only	10	60	72.5%	1.8
W_{203} `VEII`	attractors only	1	28	46.1%	2.6
W_{203} `VEII` ($N = 12$)	attractors only	1	60	39.1%	3.2

All rows at $N = 10$ unless marked, obc0. "Effective # attractors" is the participation ratio $1/\sum_i \varphi_i(x)^2$ averaged over basis states. The dissipative rules whose *weak* components proliferate behave exactly as Proposition 1 requires: every basis state has a certain fate, the conserved quantities are projectors, the charge is fixed at $t = 0$. W_{203}, with a single weak component, does not: fewer than half its basis states have a determined fate, and the fraction *falls* with N while the effective number of accessible attractors rises (max 15.0 at $N = 10$, 27.1 at $N = 12$).

The physical statement is therefore:

> With exponentially many weak components, **the initial state decides** which attractor the system reaches. With one weak component and exponentially many attractors, **the noise decides**; the initial state only sets the odds.

Both are ergodicity breaking, and both give an exponentially degenerate steady-state manifold. Only the first is Hilbert-space fragmentation.

3.4 How much memory survives, in bits

Theorem 1 says the *certain* memory does not grow. It does not say the memory is small, and it would be wrong to conclude that. Let X be a uniformly random initial basis state and T the terminal component the system ends in. The natural quantities are the entropy of the outcome $H(T)$, the noise $H(T \mid X)$ injected by the branching, and the retained information $I(X;T) = H(T) - H(T \mid X)$.

rule	regime	N	$H(T)$	$H(T \mid X)$	$I(X;T)$	ΔI per 2 sites
W_{73} `VIID`	sectors proliferate	8	3.170	0.000	3.170	—
		12	4.832	0.000	4.832	0.831
W_{109} `EIIV`	sectors proliferate	12	5.471	0.000	5.471	0.831
W_{36} `IDDV`	attractors only	12	6.588	0.710	5.878	0.889
W_{203} `VEII`	attractors only	8	3.626	0.805	2.821	—
		12	5.786	1.242	4.544	0.862
W_{219} `VEEI`	attractors only	12	4.603	1.070	3.533	0.886

All in bits, obc0, X uniform. Three things are worth reading off.

First, $H(T \mid X) = 0$ identically in the top block, which is Theorem 1 again in information-theoretic dress: when every φ_i is idempotent the outcome is a deterministic function of the input and the channel from X to T is noiseless.

Second, $H(T \mid X) > 0$ and *growing* in the bottom block — for W_{203} it rises $0.805 \to 1.024 \to 1.242$ across $N = 8, 10, 12$, an extensive loss of roughly 0.11 bits per site. A constant fraction of the capacity of this channel is consumed by noise, and that fraction does not diminish with system size.

Third, and least expected: $I(X;T)$ is *extensive in both regimes*, with strikingly similar coefficients — the increment per two sites is 0.831 for the fragmented rules and 0.862–0.889 for the multistable ones. Case B is therefore not a weak or vestigial form of memory. It stores an extensive number of bits about the initial condition, comparable in count to what genuine fragmentation stores.

The distinction survives all of this, and sharpens: it is not *how much* is remembered but *with what certainty*. In the fragmented case the map $x \mapsto T$ is a deterministic labelling, so there is a decoder that is correct with probability one and is itself a conserved quantity. In the multistable case no such decoder exists — by Theorem 1 the only zero-error questions are the WCC memberships, and there is just one weak component, so **nothing whatever about the initial state can be recovered with certainty**, even though $\sim 0.38N$ bits can be recovered on average. Fragmentation gives a noiseless code; multistability gives a noisy channel of comparable rate.

3.5 Where this sits in the current literature

Two neighbouring bodies of work make the placement precise.

Dissipative freezing. Sánchez Muñoz, Buca and co-workers [10, 11] study what individual quantum trajectories do under a *strong* symmetry: initialised in a superposition of sectors, each stochastic realisation randomly selects one and stays there, so the conservation law that holds for the density matrix is "broken" at the level of single realisations. Our case is the mirror image. There, trajectory selection happens *despite* a strong symmetry that makes the ensemble remember; here there is no strong symmetry at all, and the trajectory-level selection among ψ^N attractors is the only structure present. The two phenomena are easy to confuse from the trajectory picture alone and are distinguished exactly by whether φ_i is idempotent.

Dark states. When the attractors are single basis states — the generic case among the rules where this gap appears — the terminal components are dark states in the standard sense: configurations immune to every jump operator, which is precisely how the recurrent subspace is characterised in open reaction–diffusion settings. An exponentially large dark-state manifold reached from a structureless transient bulk is a well-recognised target for dissipative state preparation, and it is a different resource from a fragmented Hilbert space: one prepares states, the other protects them.

4 The four regimes

Writing $n_{\mathrm{wcc}} \sim a_{\mathrm{w}}^N$, $n_{\mathrm{rec}} \sim a_{\mathrm{r}}^N$, $d_{\max}^{\mathrm{rec}} \sim b_{\mathrm{r}}^N$, four regimes are possible and each has a distinct reading.

(A/D) Sectors, attractors and attractor dimension all exponential. Exponentially many superselection sectors, each retaining an exponentially large recurrent space. This is fragmentation in the full sense, and it is the regime in which the asymptotic manifold (3) can carry an exponentially large noiseless subsystem — strong fragmentation is a known route to stabilising exponentially many decoherence-free subspaces. Whether $d_{\max}^{\mathrm{rec}} = n_k m_k$ is coherence (n_k) or classical mixing (m_k) is *not* decided by the graph; see §6. In our setting the unitary rules W_{108}, W_{201}, W_{156}, W_{198} sit here together with the dissipative W_{73}, W_{109}.

(A/C) Sectors exponential, attractors trivial. Exponentially many sectors, each collapsing onto a small attractor — typically one frozen state per sector. The fragmentation is real and the memory is classical: the long-time state is a bit string labelled by the conserved charge. This is what dephasing does to a fragmented model, and it is the regime Li, Sala and Pollmann [7] reach when they couple a fragmented system to a dephasing bath: the quantum fragmentation is reduced to a classical one, the stationary state becomes separable, and what survives is classical correlations arising from fluctuations of the remaining conserved quantities. The degenerate limiting case is $W_{204} = \texttt{IIII}$, with 2^N sectors of one state each.

(B/C) Attractors exponential, sectors not, attractors trivial. The regime that provoked this report. One (or polynomially many) Krylov sectors, ψ^N dark states, a transient bulk occupying 99%+ of the Hilbert space, and non-idempotent conserved quantities. Not fragmentation: **multistability with an exponentially degenerate dark-state manifold**. The exemplar is $W_{203} = \texttt{VEII}$, whose entire 2^N-dimensional space is one weak component.

(B/D) Attractors exponential and exponentially large, sectors not. Exponentially many enclosures each exponentially large, inside a single superselection sector. This would be the most interesting regime — an exponentially degenerate manifold of exponentially large noiseless subsystems with no symmetry organising them — and it is *empty* in the 256-rule family at obc0. We record its absence as an observation, not a theorem; nothing we know forbids it, and the volume constraint $a_{\mathrm{r}} b_{\mathrm{r}} \leq 2$ leaves room for it.

5 A terminology collision worth flagging

This project now uses two independent dichotomies that are both called "strong versus weak", and they are orthogonal. Conflating them would invalidate R21's conclusions or this report's, depending on the direction of the error.

strong / weak *fragmentation* Sala *et al.* [8], Khemani *et al.* [9]	**strong / weak *symmetry*** Buča–Prosen [1]
About the *size* of the largest sector relative to the symmetry sector containing it: strong iff $D_{\max}(s)/\dim s \to 0$.	About *how* a symmetry acts: strong iff every jump operator commutes with it individually, weak iff only the generator does.
A question you ask *after* you have a sector decomposition.	A question about whether you have a sector decomposition at all.
Used in R21, which grades 108/201/156/198 as strongly fragmented.	Used here, to say what n_{wcc} counts.

A model can be weakly fragmented with a perfectly good strong symmetry (W_{60}, W_{102} in R21), and — as §3 shows — can have exponentially many steady states with no strong symmetry whatever. There is also a third, newer use of the same words in the mixed-state literature: *strong-to-weak spontaneous symmetry breaking* [18], where a strong symmetry of a density matrix degrades to a weak one and the order parameter is a fidelity correlator rather than a two-point function. That is a statement about phases of mixed states, not about sector counting, and it should not be read into either column above. Whether the boundary between our regimes A and B can be crossed as a genuine SWSSB transition in a family of channels is a natural question and one we have not touched.

6 What a population graph cannot certify

The support graph (1) is the *population* skeleton of the channel. It records which basis states feed which, and nothing about the phases. Three consequences bound every claim in this report and in the project's dissipative reports generally.

1. **It is a lower bound on fragmentation.** A channel may have conserved quantities invisible in the computational basis — the fragmentation may be quantum, i.e. in a non-product basis, in the sense of [5]. The WCC count is then an undercount. This is not hypothetical for us: R-T15 identified the single-V commutant of $W_{108}/W_{201}/W_{156}/W_{198}$ as a Temperley–Lieb algebra, and the Temperley–Lieb model is exactly the vehicle Li, Sala and Pollmann use for *quantum* HSF in an entangled basis [7]. Their fragmentation-preserving coupling produces a steady state with coherent memory carried by an extensive set of *non-commuting* conserved quantities. No diagonal graph can see any of that.

2. **It cannot split $d^{\text{rec}}_{\max}$ into $n_k \cdot m_k$.** The graph gives the dimension of an enclosure's support; (3) says that dimension factorises into a noiseless subsystem and a mixing factor, and which one carries the growth is a question about coherence. The project has answered it in both directions already: R17 found the exponential terminal SCCs to be dense, mixing digraphs rather than cycles — so m_k is large and n_k small, a classical attractor — while the X-gate correction established the opposite for permutation circuits, where the functional graph describes only populations and exponentially large decoherence-free subspaces sit underneath it. *The graph alone never settles this* — but the superoperator does, and §7 settles it.

3. **The equality** $\dim($**fixed points of** $\Phi^\dagger) = n_{\text{rec}}$ **is a classical statement.** For the population chain the conserved quantities are exhausted by the absorption probabilities (4). The full channel can carry additional fixed points built from coherences between enclosures of equal structure, which is precisely the M_{n_k} factor in (3), so n_{rec} is a lower bound on the steady-state count. *Along the diagonal the inequality runs the other way*: a diagonal operator fixed by $\Phi^\dagger$ must be harmonic *and* have no off-diagonal image, so $\dim(\mathcal{F} \cap \mathcal{D}) \leq n_{\text{rec}}$ with equality only for the monitored chain (Remark 1).

7 The asymptotic manifold, measured

Item 2 of §6 is a genuine limitation of the support graph, but it is not a limitation of the model: the split it cannot see is computable exactly at small N, and this section computes it. Doing so changes the reading of two of the rules §4 classified.

7.1 What the two factors are

Write the structure theorem (3) out with its physical content attached. Every state converges into

$$\rho(\infty) \;=\; \bigoplus_k p_k \;\; \rho_k \otimes \sigma_k, \qquad \mathcal{H}_{\text{asym}} = \bigoplus_k \mathcal{H}_{n_k} \otimes \mathcal{H}_{m_k}, \tag{5}$$

and the three objects are three different kinds of thing.

- $p_k = \mathrm{Tr}(J_k \rho_0)$ is a *classical label*, fixed at $t = 0$ and never moving again. This is the part the population graph sees.
- ρ_k is an arbitrary state on an n_k-dimensional factor. It is not damped at all; the channel acts on it as a unitary, $\rho_k \mapsto U_k \rho_k U_k^\dagger$. Everything written here survives forever. This is a *decoherence-free subspace* in the sense of Zanardi and Rasetti [14], a noiseless subsystem in the sense of Knill, Laflamme and Viola [13].
- σ_k is a *fixed* full-rank state on an m_k-dimensional factor, determined by the channel alone. Whatever is written there is erased and replaced by σ_k. It is junk.

The set of steady states is convex and compact, and its *extreme points* — the states that are not mixtures of other steady states — are read off (5) directly: p concentrated on one k, and $\rho_k = |\psi\rangle\langle\psi|$ with ψ inside a single eigenspace of U_k. Two consequences are worth stating because they are easy to get backwards.

Remark 2. An extremal steady state is usually a *mixed* state of the full system: it is $|\psi\rangle\langle\psi| \otimes \sigma_k$ and σ_k has full rank on m_k dimensions. Extremality is convex geometry, not purity.

Remark 3. There are either finitely many extremal steady states or a continuum, never anything between. If every $n_k = 1$ there are exactly K of them and the steady-state set is a simplex with K vertices — one per attractor, the classical picture. As soon as some $n_k \geq 2$ the extremals contain a copy of $\mathbb{CP}^{n_k-1}$ and the steady-state set grows a Bloch-ball face. That jump is the difference between a classical memory and a quantum one, and n_{rec} cannot detect it.

7.2 How it is computed

Exactly, with no time-stepping. The Cesàro projector at eigenvalue 1 is built from the left and right kernels of $S - \mathbb{1}$, where S is the dense 4^N one-cycle superoperator; applied to $\mathbb{1}/d$ it gives ρ_∞, whose support *is* $\mathcal{H}_{\text{asym}}$, of dimension $D = \sum_k n_k m_k$. That support is invariant, so the channel restricts to it — and the restricted channel has a *full-rank* fixed state, which is exactly the hypothesis under which the peripheral eigenoperators ($|\lambda| = 1$) of the dual map close into a $*$-algebra

$$\mathcal{A} \;=\; \bigoplus_k M_{n_k} \otimes \mathbb{1}_{m_k}, \qquad \dim \mathcal{A} = \sum_k n_k^2, \tag{6}$$

on which the dual map acts as an automorphism. Splitting $\mathcal{A}$ by its centre gives the blocks; the rank of each block's subalgebra gives n_k and the block support gives m_k.

Two points of technique. The *fixed* points of $\Phi^\dagger$ are not enough: whenever $U_k \neq \mathbb{1}$ they give only the commutant of U_k, which for any unitary rule collapses (6) to something far smaller than the truth. And the peripheral subspace must be taken from a *sorted Schur form*, not from eigenvectors: with a degenerate peripheral spectrum `numpy.linalg.eig` returns nearly parallel eigenvectors — for W_{51} at $N = 5$ all 1024 eigenvalues lie on the circle but the eigenvector matrix has numerical rank 1001 — while a Schur basis of an invariant subspace is orthonormal by construction.

Table 1: The asymptotic manifold at $N = 4$, obc0. D is its dimension, $\dim \mathcal{A} = \sum_k n_k^2$, and $\dim \mathrm{Fix}(\Phi^\dagger) < \dim \mathcal{A}$ exactly when some $U_k \neq \mathbb{1}$.

rule	word	D	$\dim \mathcal{A}$	$\dim \mathrm{Fix}$	blocks
W_{232}	DIIE	7	49	49	$M_7 \otimes \sigma_1$
W_{36}	IDDV	6	36	36	$M_6 \otimes \sigma_1$
W_{203}	VEII	4	16	10	$M_4 \otimes \sigma_1$
W_{150}	IVVI	16	256	36	$M_{16} \otimes \sigma_1$
W_{91}	VEED	3	1	1	$M_1 \otimes \sigma_3$
W_{189}	EIVE	7	2	2	$M_1 \otimes \sigma_4 \,\oplus\, M_1 \otimes \sigma_3$
W_{19}	VVVD	13	17	9	$M_4 \otimes \sigma_1 \,\oplus\, M_1 \otimes \sigma_9$

7.3 The two pure cases, and one that has both

$m_k = 1$: a pure decoherence-free subspace. $W_{232} =$ DIIE is, decoded back to an elementary cellular automaton, $f(l, s, r) =$ majority. Its asymptotic algebra is the *full* matrix algebra M_7 with $U_k = \mathbb{1}$: nothing is erased, the asymptotic manifold is every state on a seven-dimensional subspace, and the extremal steady states form a continuum $\mathbb{CP}^6$. The population graph reports seven fixed points and calls them seven classical pointer states; the truth is $\log_2 7 \approx 2.8$ *qubits* held exactly. This reproduces, from the superoperator, the count $\dim \mathcal{A} = n_{\mathrm{rec}}^2 = 49, 121$ at $N = 4, 5$ that `permutation/coherence.py` obtains by an unrelated route (label synchronisation along the orbit), which is a useful cross-check on both.

$n_k = 1$: pure mixing. $W_{91} =$ VEED has a single block with $n = 1$, $m = 3$: one steady state, of rank three, with spectrum exactly $(\frac{2}{5}, \frac{2}{5}, \frac{1}{5})$, reached from every initial state. No memory of any kind; the manifold is a point. At $N = 5$ the manifold is a single *pure* state ($D = 1$), which is the $N \equiv 2 \pmod 3$ case of the closed form for W_{91}'s largest recurrent class recorded in this project's notes. $W_{189} =$ EIVE has two such blocks, $(1, 4)$ and $(1, 3)$: exactly two extremal steady states, both mixed, and the manifold is the line segment joining them. That is bistability in its purest form — one classical bit survives, no qubit does — and it matches $n_{\mathrm{wcc}}(W_{189}) = 2$.

Both at once. $W_{19} =$ VVVD carries one block of each kind, $\mathbb{C}^4 \oplus (\mathbb{C}^1 \otimes \mathbb{C}^9)$. Its decoherence-free block *rotates*: $\dim \mathrm{Fix} = 8 < 16$ and the peripheral phases are $\{0, \pi\}$, so U_0 has two doubly degenerate eigenvalues of opposite sign and the encoded state flips every cycle. The other block has a unique rank-nine σ. The extremal steady states are two Bloch spheres plus one isolated point.

7.4 Two findings

1. **No block mixes the two factors.** Across all 256 rules at $N = 4$ and again at $N = 5$, obc0, not one block has $n_k > 1$ *and* $m_k > 1$. Every block is either a pure decoherence-free subspace or a single state. The genuinely quantum intermediate case — a noiseless *subsystem*, where information hides in a tensor factor and is invisible to any subspace projection — does not occur in this family at these sizes. At $N = 4$: 141 rules have some $n_k > 1$, 49 have some $m_k > 1$, four rules ($W_{19}, W_{61}, W_{103}, W_{179}$) carry both kinds of block in the same channel, and 70 rules have every block trivial, of which 68 have a unique steady state. At $N = 5$ the counts move (136 and 57) but the structure does not: still no mixed block, and still the same four rules carrying both kinds. We record this as an observation, not a theorem; it is exactly the kind of statement that could fail at larger N, and 512 decompositions are not many.

2. **Two of the "multistable" rules are not.** W_{203} and W_{36} are among the eight rules that motivated this report: $n_{\mathrm{wcc}} = 1$ and $n_{\mathrm{wcc}} = 10$ against $n_{\mathrm{rec}} = 28$ and 60 at $N = 10$, with certain fractions of 46% and 73% (§3.3). Their asymptotic algebras are M_4 and M_6 at $N = 4$, and M_5 and M_9 at $N = 5$ — *single* blocks. The channel's asymptotic evolution on them is unitary; there are no competing attractors for noise to choose between. What §3.3 measured is a property of the *monitored* chain, in which the computational basis is dephased once per cycle, and that is the right object for the questions §§3–3.4 ask. But the sentence "the noise decides which attractor" is a statement about the monitored channel and must not be carried over to the unmonitored one. The distinction is the same one R7 drew for the pair graph and the X-gate correction drew for permutation circuits; it now has numbers on both sides.

8 Memory, and what it costs to correct errors

8.1 The asymptotic algebra is the code

Equation (6) is not merely analogous to an error-correcting code; it is the same object. Blume-Kohout, Ng, Poulin and Viola [12] proved that every *information-preserving structure* of a quantum process — and their notion subsumes noiseless subsystems, decoherence-free subspaces, pointer bases and error-correcting codes — is a matrix algebra, and is isometric to the fixed-point algebra of an associated unital process. What our $\mathcal{A}$ is, is that algebra for the channel iterated forever. So the R22 counts sort into three distinct kinds of resource:

quantity	what it counts	code-theoretic name
$n_{\mathrm{wcc}} = \dim(\mathcal{C} \cap \mathcal{D})$	protected classical bits	zero-error capacity
n_k in (6)	protected qubits	decoherence-free subspace
$n_{\mathrm{rec}} - n_{\mathrm{wcc}}$	nothing	—

The third row is Theorem 1 restated in the language of codes. Extra terminal components inside one weak component add conserved quantities, but they add no conserved *projections*, hence no distinguishable-with- certainty codewords, hence nothing to the zero-error capacity. In the same language, §3.4's finding is that the ordinary (average-case) capacity is extensive in both regimes: what multistability destroys is *zero-error* capacity specifically.

It is worth recording the operator-level statement behind that, which is stronger than the diagonal version proved in Theorem 1.

Proposition 6 (Every conserved question is a strong symmetry)**.** *Let $P = P^2 = P^\dagger$ satisfy $\Phi^\dagger(P) = P$. Then $[P, K_j] = 0$ for every j. Consequently $\mathcal{C}$ and $\mathcal{F}$ contain exactly the same projections, and the entire gap $\mathcal{F} \setminus \mathcal{C}$ consists of non-idempotent elements.*

Proof. For ρ supported in $\operatorname{ran} P$ we get $\operatorname{Tr}(P\,\Phi(\rho)) = \operatorname{Tr}(\Phi^\dagger(P)\rho) = \operatorname{Tr}(P\rho) = 1$, so $\operatorname{ran} P$ is invariant. The same computation with $\mathbb{1} - P$, which is also a conserved projection, makes $\operatorname{ran} P^\perp$ invariant. Every K_j therefore preserves both summands of $\mathcal{H} = \operatorname{ran} P \oplus \operatorname{ran} P^\perp$, i.e. is block-diagonal, i.e. commutes with P. □

This also explains why $\mathcal{F}$ is not an algebra while $\mathcal{C}$ is. A commutant is closed under products by construction; $\mathcal{F}$ is closed under products only when $\Phi^\dagger$ is multiplicative on it, and $\Phi^\dagger$ is a $*$-automorphism precisely in the closed case $\Phi^\dagger(A) = U^\dagger A U$. That is §1 seen from the algebraic side. The general criterion is Frigerio's [15]: with a *faithful* stationary state, $\mathcal{F}$ is a von Neumann algebra and coincides with $\mathcal{C}$. A large transient region is exactly what forbids a faithful stationary state, so our dissipative rules sit as far from that hypothesis as possible — measured at $N = 4$, obc0, $(\dim \mathcal{C}, \dim \mathcal{F})$ is $(30, 49)$ for W_{232}, $(20, 36)$ for W_{36} and $(1, 10)$ for W_{203}: one strong symmetry, ten conserved quantities.

8.2 Fragmentation as a source of logical qubits: Iadecola's construction

Iadecola [16] asks what fragmentation buys when it is combined with a symmetry rather than considered alone. In a charge-conserving, kinetically constrained chain, the projectors onto the Krylov sectors and the generators of charge conjugation and translation generate a *non-Abelian* algebra, and the consequence is exponentially many logical qubits encoded in degenerate pairs of eigenstates, which may be highly entangled.

The mechanism translates into our language cleanly, and the translation makes the difference visible. His degeneracies come from a symmetry that *maps one fragment onto another*, so that the pair carries a two-dimensional irreducible representation. In (6) that is precisely $n_k = 2$: a non-Abelian commutant is the same thing as a block of dimension greater than one, and this project has already met one — R-T15 identified the single-V commutant of $W_{108}/W_{201}/W_{156}/W_{198}$ as non-Abelian, of Temperley–Lieb type.

Two differences matter.

1. **Closed versus open.** Iadecola's setting is Hamiltonian, so there is no transient part and $\mathcal{C} = \mathcal{F} = \mathcal{A}$; the distinction this whole report is about does not arise. Ours has transients, and they break the identification: W_{203} at $N = 4$ has $\dim \mathcal{C} = 1$ — no strong symmetry whatsoever — and yet $\mathcal{A} = M_4$, a four-dimensional decoherence-free subspace. **A decoherence-free subspace need not be protected by any symmetry of the dynamics**; it can be a property of the recurrent set alone, invisible in $\mathcal{C}$ because $\mathcal{C}$ must also respect the transients.

2. **Where the coherence comes from.** His comes from a symmetry relating sectors. Ours comes from the resets acting *trivially* on the recurrent set: a reset destroys $|x\rangle\langle y|$ only if it fires on a bit where x and y differ, so once the dynamics has settled onto configurations the resets never touch, all coherences among them survive. That is the J-label criterion of the X-gate correction, and it is why W_{232}'s algebra is the whole of $M_{n_{\mathrm{rec}}}$.

A third point of contact is worth flagging as a caution rather than a result. Iadecola derives from the same algebra *necessary conditions* for the usual experimental signatures of fragmentation, such as the late-time persistence of density imbalances. Our §3.3 numbers are exactly such late-time observables, and Finding 2 of §7 is an instance of his warning: a late-time imbalance measured in the computational basis reports on the monitored chain, and can indicate multistability where the unmonitored channel has a single coherent block.

8.3 The majority rule as an autonomous memory

Guedes, Winter and Müller [17] propose exactly this family of automata as measurement-free quantum error correction: a QCA whose local update is a classical density-classification rule, run as a decoder for a repetition code, with no syndrome extraction anywhere. The two rules they study are Wolfram's rule 232 — local majority voting — and Toom's two-line voting in two dimensions. Rule 232 is *our* W_{232}: decoding the gate word DIIE through the project's encoding gives $(c_p, c_q) = (f(l, 0, r), f(l, 1, r))$ with $f =$ majority, so the automaton whose asymptotic algebra Table 1 reports as M_7 is the one they run as a decoder.

That makes the comparison direct, and it is not flattering to fragmentation.

Three readings of Table 2.

The automaton corrects exactly one error, at every size. A single flipped bit is an isolated minority and majority voting removes it. Two *adjacent* flipped bits are not: each of them sees one neighbour agreeing, so the pair is stable, and it is stable because it is one of the automaton's own protected pointer states. The autonomous distance is therefore 1 for every N, against a repetition-code distance of $\lfloor (N-1)/2 \rfloor$. This is precisely the "islands" failure Guedes

Table 2: W_{232} = majority voting as an autonomous memory, obc0. d is the largest weight such that *every* error of that weight or less is undone by the automaton itself. The obc0 boundary pins the left and right ends to 0, so the all-ones codeword is unprotected at its edges and $d(1^N) = 0$; the periodic counterpart is deliberately not run here.

N	attractors	basin 0^N	basin 1^N	code basin	$d(0^N)$	repetition d
8	44	41	12	20.70%	1	3
10	117	99	29	12.50%	1	4
12	305	239	70	7.54%	1	5
14	798	577	169	4.55%	1	6
16	2091	1393	408	2.75%	1	7
18	5473	3363	985	1.66%	1	8
20	14328	8119	2378	1.00%	1	9

et al. identify, and R22's vocabulary names its cause: the islands are not decoding failures, they are *other weak components*, and the automaton preserves them with the same fidelity with which it preserves the codewords. The very property that makes W_{232} a good passive memory — $\mathcal{A} = M_{n_{\mathrm{rec}}}$, everything protected — is what makes it a bad decoder.

The code basin shrinks exponentially. The two codeword basins are Pell and NSW numbers, growing like $(1+\sqrt{2})^{N/2} \approx 1.554^N$ against a state space of 2^N, so the fraction of configurations the automaton returns to a codeword decays like 0.777^N: half the space at $N = 4$, one percent at $N = 20$. Meanwhile the attractor count grows like φ^N. A decoder is useful when the intended codewords have basins of measure approaching one; here they have measure approaching zero.

Hence: autonomous error correction wants the *opposite* of fragmentation. The figure of merit for a self-correcting memory is a *small* asymptotic algebra with *large* basins — ideally $K = 2$ blocks, or one block with $n_k = 2$, and a transient region covering essentially the whole space. Exponential n_{wcc} means exponentially many competing enclosures, so the intended codeword's basin is an exponentially small fraction of the space and the automaton faithfully preserves errors. This is the same obstruction that sends Guedes et al. to Toom's rule in two dimensions, where the two consensus states have basins of measure approaching one. Fragmentation is a good *passive* memory — it protects what is already there — and a bad *active* one, because it protects the wrong states just as well.

The absorption probabilities are the decoder. One further identification makes §3.3 directly usable here. If the intended codeword is terminal class i and an error takes the state to x, then $\varphi_i(x)$ of (4) is exactly the probability that the automaton restores the codeword: φ *is* the decoder's success landscape, and the "certain fraction" reported in §3.3 is the fraction of configurations decoded with probability one. Read that way, the 100% for W_{73}/W_{109} against 46% for W_{203} is a statement about decoders, and Theorem 1 says the certain part is supported on whole weak components and nowhere else.

9 Summary

- WCCs are the blocks of a *strong symmetry*: the finest basis decomposition making every Kraus operator block-diagonal (Prop. 3), and $n_{\mathrm{wcc}} = \dim(\mathcal{C} \cap \mathcal{D})$ exactly (Prop. 4), which is the quantity Moudgalya–Motrunich's criterion is stated in terms of. Exponential growth of it is Hilbert-space fragmentation.

- Terminal SCCs are *enclosures*: supports of extremal steady states, with $n_{\rm rec} = \dim(\mathcal{F}\cap\mathcal{D})$ (Prop. 5). The familiar inequality $n_{\rm rec} \geq n_{\rm wcc}$ is the algebra inclusion $\mathcal{C} \subseteq \mathcal{F}$.

- Exponentially many terminal SCCs inside one WCC is **not** fragmentation. The projections in $\mathcal{F}$ are exactly the unions of weak components (Thm. 1), so the idempotent part of the conserved-quantity space cannot see the extra attractors at all; the attractor is then selected by noise rather than fixed by the initial state — a 100% versus 46% effect in direct measurement (§3.3).

- But the memory is not weak, only uncertain: $I(X;T)$ is extensive in both regimes at comparable rates (§3.4). Fragmentation is a noiseless code; multistability is a noisy channel of similar capacity whose zero-error capacity is nil.

- The gap requires the channel to be **both non-unital and branching**: unital channels have no transient part (Prop. 1) and functional graphs have one attractor per component (Prop. 2). Hence no closed-system analogue, and in this family all eight rules exhibiting the gap carry exactly one V together with at least one reset.

- "Strong/weak" means two different things in this project, and now a third thing in the mixed-state literature (§5).

- Everything in §§1–5 is certified at the level of populations only (§6); coherent fragmentation would be invisible, and our own Temperley–Lieb structure is a live candidate for it.

- §7 removes the second of those limitations by exact computation. Every enclosure splits as $M_{n_k} \otimes \sigma_{m_k}$, and across all 256 rules at $N = 4$ and $N = 5$ **no block has both factors non-trivial**: each is a pure decoherence-free subspace or a single state. W_{232} holds $\log_2 7$ qubits exactly at $N = 4$; W_{91} holds nothing; W_{189} holds one classical bit; W_{19} holds both kinds at once. Crucially W_{203} and W_{36}, filed as multistable in §4, have *single* blocks. For W_{203} that does settle it — "the noise decides" is true of the monitored chain and false of the channel, at $N = 4$ and $N = 5$ alike. For W_{36} it does not: a single block still leaves $\dim(\mathcal{F} \cap \mathcal{D}) = 9$ against $n_{\rm wcc} = 7$ at $N = 5$, so its gap is real. R24 separates the two cases across the family.

- A decoherence-free subspace need not come from a symmetry: W_{203} at $N = 4$ has $\dim \mathcal{C} = 1$, no strong symmetry at all, and $\mathcal{A} = M_4$. In the closed-system setting of Iadecola's symmetry fragmentation [16] the two coincide; transients separate them.

- Read as codes (§8): $n_{\rm wcc}$ is zero-error classical capacity, n_k is protected qubits, and the gap is nothing. Every conserved projection is a strong symmetry (Prop. 6), so $\mathcal{C}$ and $\mathcal{F}$ differ only in non-idempotents.

- Autonomous error correction wants the *opposite* of fragmentation. W_{232} is Guedes–Winter–Müller's majority-voting decoder; its islands are other weak components, protected exactly as well as the codewords, so the autonomous distance is 1 at every N while the codeword basin decays like 0.777^N.

Reproduce

The measurement of §3.3 needs transition *probabilities*, which the project's component engine does not carry: `graph/scc.py` works with the unweighted successor relation, since that is all that component structure requires. The weighted one-cycle chain is therefore rebuilt from the gate semantics (I keeps, $V = H$ splits $\frac{1}{2}/\frac{1}{2}$, D and E reset), in `scaling/absorption.py`, at cost $O(4^N)$. It is a small-N diagnostic, not a sweep.

```
python -m qca_fragmentation.scaling.absorption --table
pytest qca_fragmentation/tests/test_absorption.py    # 101 tests
```

The decomposition of §7 needs the coherences too, so it works with the dense 4^N superoperator of `quantum/peripheral.py`; the splitting itself lives in `quantum/asymptotic.py`, which also carries the basin and autonomous-distance measurements of §8.

```
python -m qca_fragmentation.quantum.asymptotic --table
python -m qca_fragmentation.quantum.asymptotic --memory 232
python -m qca_fragmentation.quantum.asymptotic --census 4
pytest qca_fragmentation/tests/test_asymptotic.py   # 69 tests
```

The censuses are stored as `analytics/asymptotic_blocks_obc0_N4.json` and its $N = 5$ counterpart, one entry per rule, each carrying its own consistency flag ($\sum_k n_k m_k = D$ and $\sum_k n_k^2 = \dim \mathcal{A}$); all 512 pass. The tests pin Table 1 block by block, check seed-independence of the centre splitting, confirm that unitary rules lose nothing (Prop. 1 seen from the manifold side), and pin Table 2 including the constancy of the autonomous distance.
The tests of `scaling/absorption.py` pin every number quoted in §3.3 and §3.4; check both directions of Theorem 1, including the idempotent counts 19/19, 9/28 and 0/13 and the bound $\#\{\text{idempotents}\} \leq n_{\text{wcc}}$; verify the double stochasticity of Proposition 1 (holding for the unitary rules, failing for the dissipative ones); and confirm Proposition 2 on the V-free rules. The extensivity of $I(X;T)$ is checked as an equality of successive increments to 5%.

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